



Figure 2: The App Bundle format – APK is generated with only the required resources

an application, as per their needs. So, initially, a base model of the application will be given, after which the user can download and upgrade the additional features, when required.

How Android App Bundle works

Android delivers the APK with the required resources using what is called the split APK mechanism. This is available for Android Lollipop devices and higher versions. Google Play uses this mechanism to split the large app into a smaller APK, as per the target device's requirements.

According to Google, there are three types of split APKs —the base APK, the configuration APK and the dynamic feature APK.

- The base APK contains the basic requirements for the application. This is the mandatory first APK to be installed.
- The configuration APK includes the specific data, based on the device's requirements. The configuration APK is generated by Google Play from the App Bundle that is uploaded to the store. It automatically allocates the resources such as CPU architecture, language, etc, from the set of code provided.
- The dynamic feature APK has the optional features, which can be installed as required by the user.

Figure 3 shows the split APK mechanism — for a basic device and a more resource-specific device. Here, B means 'Basic APK', C stands for 'Configuration APK', and D for 'Dynamic feature APK'.

Deploying the Android App Bundle

The Android App Bundle is in the Beta stage and, hence, to use the feature, we need to download the Android Studio 3.2 Preview Version. (*Warning: This is not a stable*

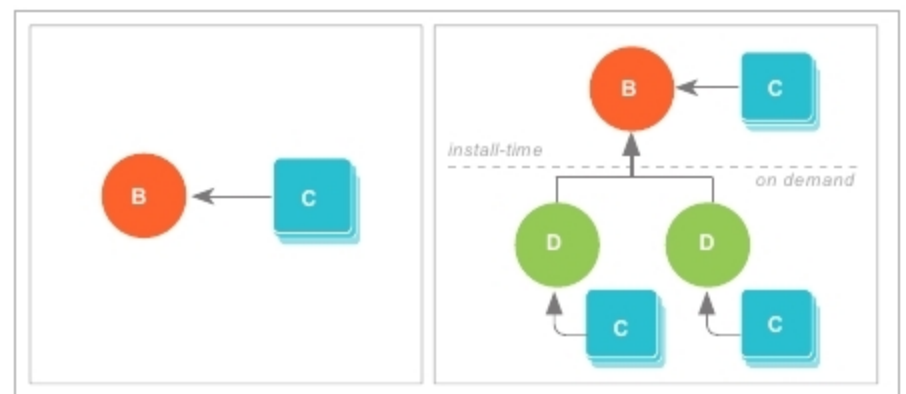


Figure 3: On the left is a simple app with a base APK (B) and some configuration APKs (C). On the right is a more complex app with two dynamic feature APKs (D) and the corresponding configuration APKs (C) for download on demand

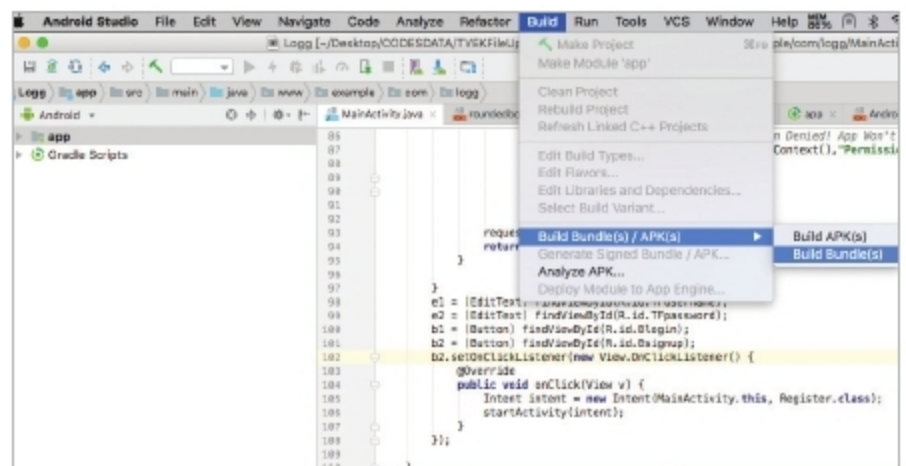


Figure 4: Building bundles

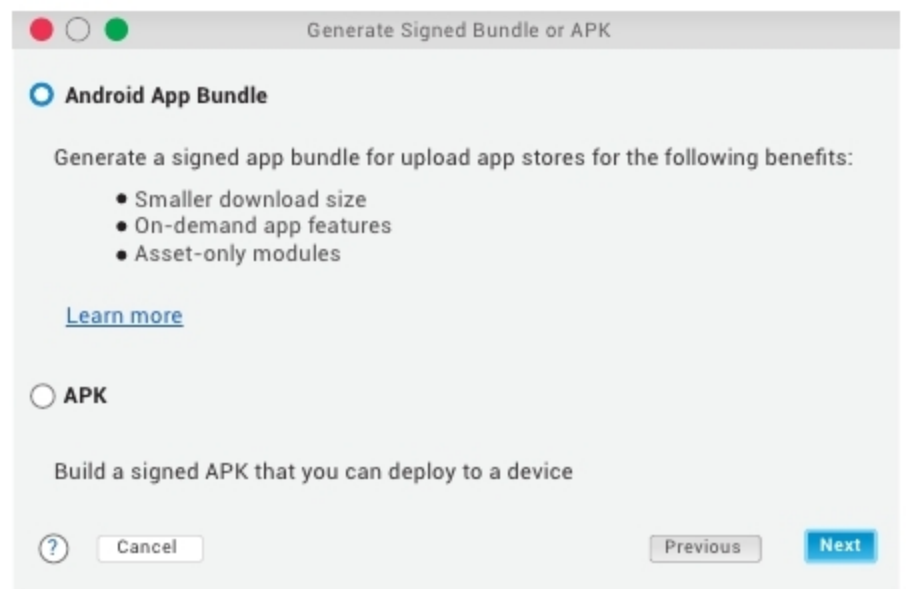


Figure 5: Generate Signed Bundle

version.) After downloading, open Android Studio and import your code. Select the *Build* menu, and from there, choose *Build Bundles* to do so.

To upload the bundle to Play Store, select the *Generate Signed Bundle/APK(s)* option from the *Build* menu (Figure 5).

Now, select *Keystore* and provide your credentials. This is similar to APK signing. The final bundle can be found in the *.aab* format, to upload to Play Store. **END** 

By: Jis Joe Mathew

The author is an assistant professor at Amal Jyothi College of Engineering, Kanjirapally, Kerala. He can be contacted at jisjoemathew@gmail.com.